

Information Security - Important Questions with Detailed Answers

Prepared from the uploaded Unit I and Unit II notes. Suitable for 5-mark and 8-mark university examinations.

1. Different Types of Cryptographic Attacks with Examples

Cryptographic attacks are classified into Passive and Active attacks.

Passive Attacks:

- Eavesdropping – attacker secretly listens to communication. Example: Sniffing passwords on open Wi-Fi.
- Traffic Analysis – attacker studies communication patterns without reading data.

Active Attacks:

- Masquerade – attacker pretends to be an authorized user.
- Replay Attack – previously captured messages are retransmitted.
- Message Modification – attacker changes message contents.
- Denial of Service (DoS) – attacker floods the server making services unavailable.

2. Compare DES and AES

Feature	DES	AES
Year	1977	2001
Block Size	64 bits	128 bits
Key Size	56 bits	128/192/256 bits
Rounds	16	10/12/14
Structure	Feistel	SPN
Security	Weak	Very Strong
Speed	Slower	Faster

AES replaced DES because DES has a short 56-bit key and is vulnerable to brute-force attacks.

3. Strengths and Weaknesses of Classical Encryption

Strengths:

- Easy to understand.
- Fast for small messages.
- Low computational cost.
- Good for learning cryptography.

Weaknesses:

- Small key space.
- Vulnerable to frequency analysis.
- Easily broken by brute-force attacks.
- Not suitable for modern communication.

4. Types of Digital Security Controls

Administrative Controls:

- Security policies
- User awareness training
- Incident response planning

Technical Controls:

- Encryption (AES)
- Firewalls
- IDS/IPS
- MFA
- Digital signatures

Physical Controls:

- CCTV
- Biometric access
- Locked server rooms
- Fire suppression
- Power backup

5. Difference Between Cyber Security and Cloud Security

Cyber Security protects computers, servers, networks and information systems from cyber attacks.

Cloud Security protects cloud-hosted applications, services and data. It uses IAM, encryption, virtual firewalls, MFA and follows the shared responsibility model.

6. Modes of Operation of Block Cipher

ECB: Encrypts each block independently. Simple but insecure.

CBC: XORs plaintext with previous ciphertext. Uses Initialization Vector (IV). More secure.

CFB: Converts block cipher into a stream cipher using ciphertext feedback.

OFB: Uses encryption output as feedback. Suitable for noisy channels.

7. Explain Types of Digital Security Controls with Examples

Administrative Example: Password policy, employee training.

Technical Example: Firewall, AES encryption, Multi-Factor Authentication.

Physical Example: CCTV cameras, biometric scanners, server room locks.

8. Difference Between Detective and Corrective Controls

Detective Controls:

Purpose: Detect security incidents.

Examples: IDS, Log monitoring, SIEM, Antivirus alerts.

Corrective Controls:

Purpose: Recover after an incident.

Examples: Patch management, Backup restoration, Malware removal, System recovery.

Exam Tips

- Draw a comparison table for DES vs AES.
- Mention examples for every control type.
- Write advantages and disadvantages wherever possible.
- Underline keywords such as Confidentiality, Integrity, Availability, Encryption, Firewall, MFA and IDS/IPS.